Lab Sheet 4

Database Management System Lab (BTCS203P)

B. TECH AI&ML, CYBER SECURITY, Data Science 3RD Semester (Session 2026-2025)

Objective:

To understand and practice SQL Operators.

Prerequisites:

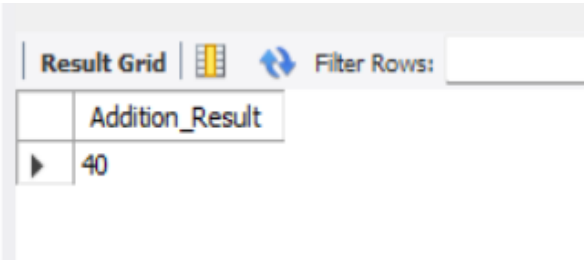
1. Basic knowledge of SQL.
2. MySQL installed and set up.

Lab Sheet 4: SQL Operators

Arithmetic Operators

1. Perform addition of two numbers using +.

```
SELECT 25 + 15 AS Addition_Result;
```

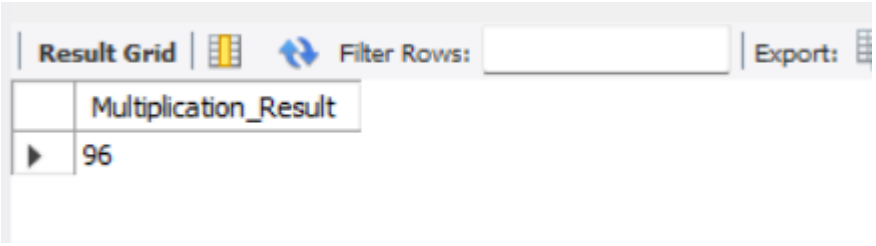


2. Perform subtraction using -.

```
SELECT 50 - 18 AS Subtraction_Result;
```

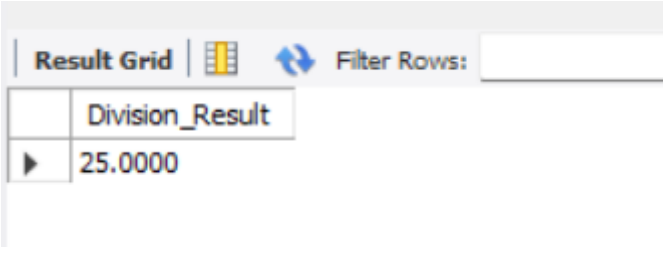
3. Perform multiplication using *.

```
SELECT 12 * 8 AS Multiplication_Result;
```



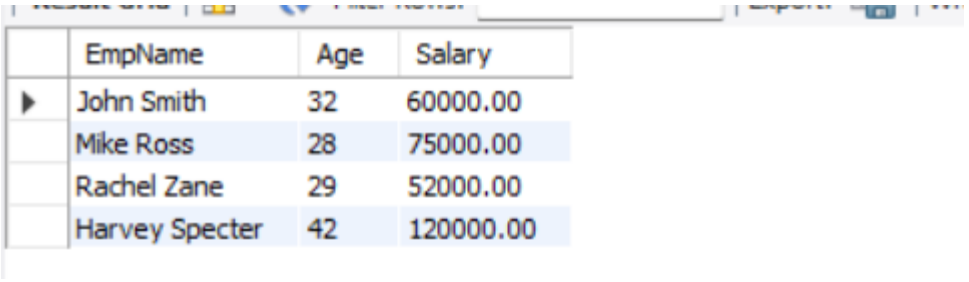
4. Perform division using /.

```
SELECT 100 / 4 AS Division_Result;
```



5. Calculate the total amount as Price × Quantity.

```
SELECT 49.99 * 5 AS Total_Amount;
```

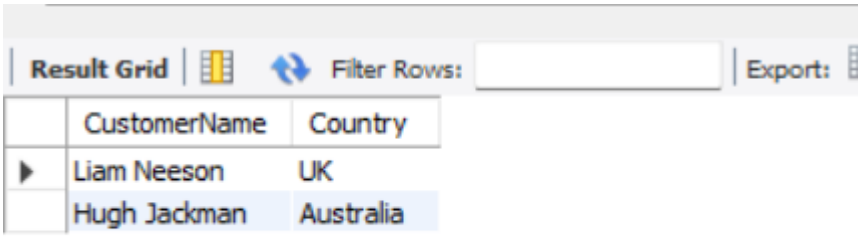


EmpName	Age	Salary
John Smith	32	60000.00
Mike Ross	28	75000.00
Rachel Zane	29	52000.00
Harvey Specter	42	120000.00

Logical Operators

6. Demonstrate AND with two conditions.

```
SELECT EmpName, Age, Salary
FROM Employees
WHERE Age > 25 AND Salary > 50000.00;
```



Result Grid | Filter Rows: | Export:

CustomerName	Country
Liam Neeson	UK
Hugh Jackman	Australia

7. Demonstrate OR with two conditions.

```
SELECT ProductName, Category, Price
FROM Products
WHERE Category = 'Electronics' OR Category = 'Appliances';
```

8. Demonstrate NOT with a condition.

```
SELECT CustomerName, Country
FROM Customers
WHERE NOT Country = 'USA';
```

9. Find students belonging to a specific course AND city.

```
SELECT StudentName, Course, City
FROM Students
WHERE Course = 'Computer Science' AND City = 'New York';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

StudentName	Course	City
Alice	Computer Science	New York
Ethan	Computer Science	New York

10. Find employees belonging to either of two departments.

```
SELECT EmpName, Department
FROM Employees
WHERE Department = 'Sales' OR Department = 'Marketing';
```

Comparison Operators

11. Demonstrate the = operator.

```
SELECT EmpName, JobTitle, Department
FROM Employees
WHERE JobTitle = 'Manager';
```

Result Grid | Filter Rows: | Export: | Wrap Cel

EmpName	JobTitle	Department
John Smith	Manager	Sales
Harvey Specter	Manager	Legal

12. Demonstrate the > operator.

```
SELECT ProductName, Price
FROM Products
WHERE Price > 100.00;
```

13. Demonstrate the < operator.

```
SELECT ProductName, StockQuantity
FROM Products
WHERE StockQuantity < 10;
```

Result Grid | Filter Rows: | Export:

ProductName	StockQuantity
Blender	8
Desk (	5

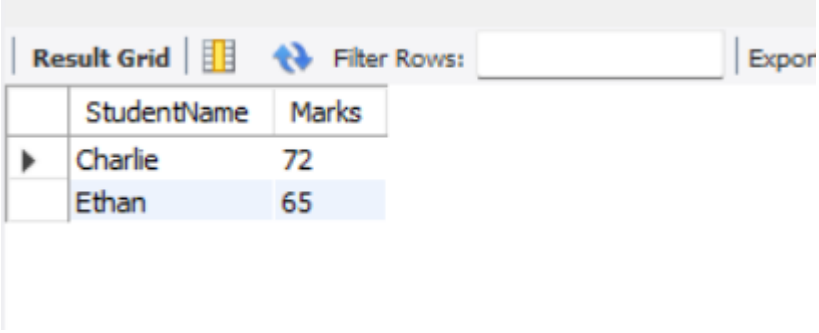
Blender

14. Demonstrate >= and <=.

```
SELECT StudentName, Marks
FROM Students
WHERE Marks >= 60 AND Marks <= 80;
```

15. Demonstrate the <>/!= operator.

```
SELECT EmpName, Status
FROM Employees
WHERE Status <> 'Terminated';
```

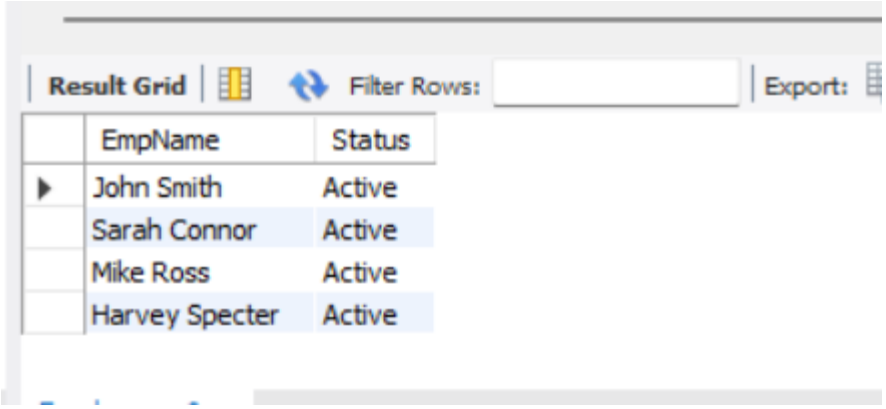


StudentName	Marks
Charlie	72
Ethan	65

Special Operators

16. Demonstrate BETWEEN for salary range.

```
SELECT EmpName, Status
FROM Employees
WHERE Status != 'Terminated';
```



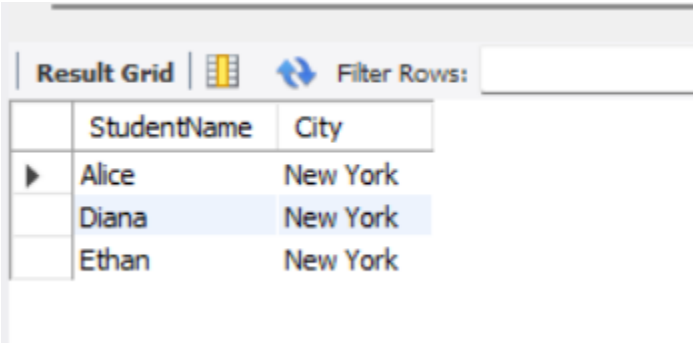
EmpName	Status
John Smith	Active
Sarah Connor	Active
Mike Ross	Active
Harvey Specter	Active

17. Demonstrate IN for selecting multiple cities.

```
SELECT StudentName, City
FROM Students
WHERE City IN ('New York', 'Chicago', 'Boston');
```

18. Demonstrate NOT IN.

```
SELECT StudentName, City
FROM Students
WHERE City NOT IN ('Chicago', 'Dallas');
```



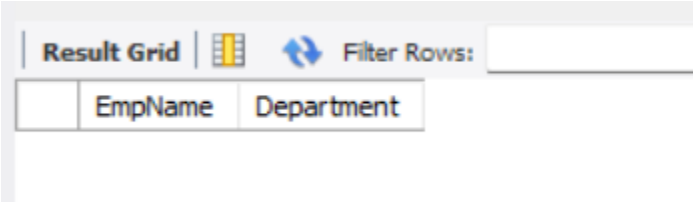
Result Grid | Filter Rows:

StudentName	City
Alice	New York
Diana	New York
Ethan	New York

19. Demonstrate LIKE for names beginning with a particular letter.

20. Demonstrate IS NULL and IS NOT NULL.

```
SELECT EmpName, Department
FROM Employees
WHERE Department IS NULL;
SELECT EmpName, Salary
FROM Employees
WHERE Salary IS NOT NULL;
```



Result Grid | Filter Rows:

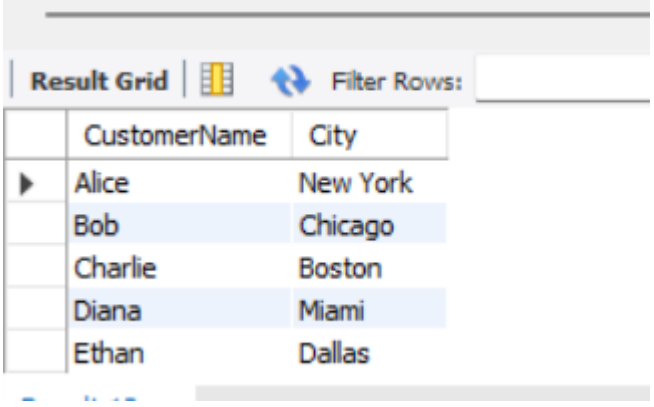
EmpName	Department

USE LabSheet4DB;

Set Operations

21. Demonstrate UNION using two compatible queries.

```
SELECT CustomerName, City FROM NorthRegion_Customers
UNION
SELECT CustomerName, City FROM SouthRegion_Customers;
```



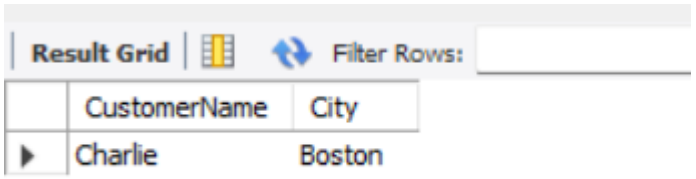
Result Grid | Filter Rows:

CustomerName	City
Alice	New York
Bob	Chicago
Charlie	Boston
Diana	Miami
Ethan	Dallas

22. Demonstrate UNION ALL.

23. Demonstrate INTERSECT where supported by the DBMS.

```
SELECT CustomerName, City FROM NorthRegion_Customers
INTERSECT
    SELECT CustomerName, City FROM SouthRegion_Customers;
```

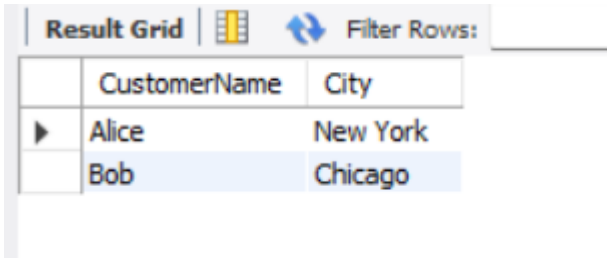


Result Grid | Filter Rows:

	CustomerName	City
▶	Charlie	Boston

24. Demonstrate EXCEPT/MINUS where supported.

```
SELECT CustomerName, City FROM NorthRegion_Customers
EXCEPT
    SELECT CustomerName, City FROM SouthRegion_Customers;
```

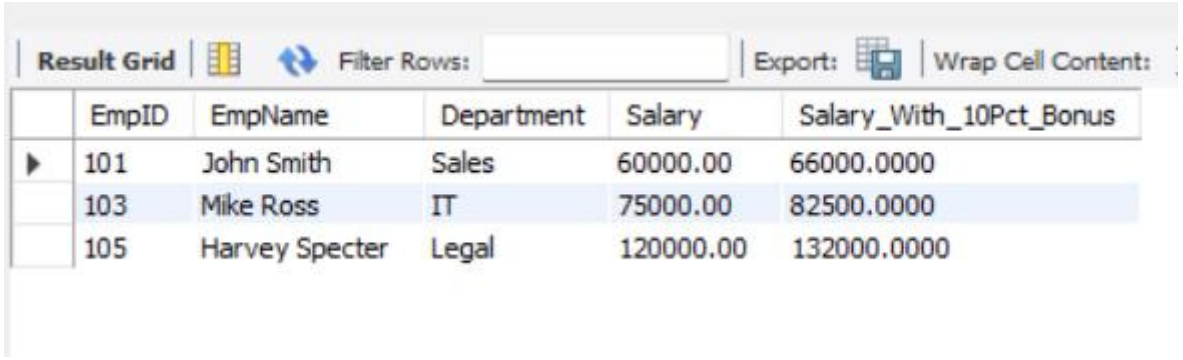


Result Grid | Filter Rows:

	CustomerName	City
▶	Alice	New York
	Bob	Chicago

25. Perform a suitable problem using multiple operators together, such as Arithmetic + Logical + Comparison + Special operators.

```
SELECT
   EmpID,
   EmpName,
   Department,
   Salary,
   (Salary * 1.10) AS Salary_With_10Pct_Bonus
FROM Employees
WHERE
   Status = 'Active'
   AND (Salary * 1.10) > 55000.00
   AND Age BETWEEN 25 AND 45
   AND Department IS NOT NULL
   AND EmpName NOT LIKE 'R%';
```



EmpID	EmpName	Department	Salary	Salary_With_10Pct_Bonus
101	John Smith	Sales	60000.00	66000.0000
103	Mike Ross	IT	75000.00	82500.0000
105	Harvey Specter	Legal	120000.00	132000.0000